

HIMANSHU PAITHANE

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EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY

Master of Science, Computer Science (GPA: 3.9/4.0)

May 2026

WORK EXPERIENCE

Data Engineer, ML Analytics | Panasonic North America

May 2025 - May 2026

- Developed and deployed an **Azure Databricks** analytics pipeline to transform and reconcile **300,000+** customer records using an NLP and fuzzy matching algorithm to improve customer segmentation, data quality, and the reliability of downstream analytics and business decision-making.
- Built a scalable **SQL and PySpark** workflow to automate data validation, schema checks, ETL reconciliation and data quality monitoring while improving pipeline reliability and ensuring trusted datasets for analytics products, dashboards and machine learning models.
- Partnered with Panasonic's B2B Marketing team to develop an analytics platform integrating **GA4** and **BigQuery** to deliver curated datasets and standardized KPI definitions that powered self-service analytics through Power BI dashboards which helped improve stakeholder decision-making.
- Managed Databricks notebooks and Unity Catalog assets to organize analytics workflows and improve discoverability of shared datasets across multiple teams.

Data Scientist | Jar (Fintech)

Mar 2024 - Aug 2024

- Developed, validated and deployed an **ML-based** affluence scoring model using **LightGBM** and **XGBoost** to classify new users into spending-affinity tiers, improving customer targeting decisions while replacing a third-party solution that saved **~\$15k/month**.
- Built Airflow-orchestrated **ETL** pipelines to process high-volume customer transaction events generated by **~20k** daily active users streamed through Kafka, transforming raw data stored in **AWS S3** into curated **MongoDB** datasets that powered analytics, KPI reporting, product insights and machine learning models.
- Applied **NLP** techniques to transform **terabytes** of unstructured financial transaction SMS data into structured datasets for user segmentation, behavioral analysis, spending-trend identification, and predictive modeling, with insights delivered through interactive **Tableau** dashboards.

Data Science Intern | SymphonyAI (B2B SaaS)

May 2023 - Aug 2023

- Built and optimized machine learning models for **time-series forecasting** and pattern recognition of motor health to identify exceptions and risk factors using sensor-derived time-series data, improving accuracy by **23%** through feature engineering, hyperparameter tuning, performance analysis and model evaluation.
- Improved accuracy of production ML models including Regression based models and **neural networks** through feature engineering, statistical analysis, and hyperparameter tuning, boosting **F1-score to 91%** and enabling more reliable predictions at scale.
- Built end-to-end data preprocessing, feature engineering, model training, and validation workflows across diverse ML models and pipelines to deliver reliable, production-oriented predictions and performance analysis.

Data Science Intern | Sharp

Jun 2021 - Sep 2021

- Built end-to-end machine learning workflows using **Python** libraries including NumPy, pandas, PyTorch, scikit-learn and TensorFlow to deliver high-accuracy, production-ready predictive models for maintenance scheduling and device performance.
- Designed, executed, and analyzed large-scale **A/B tests** to evaluate product features and business strategies, applying statistical methods to generate actionable insights that directly guided product decisions and user engagement improvements.
- Optimized distributed data pipelines using **Apache Spark & Hadoop** to process large datasets, boosting efficiency by 4x and enabling scalable analytics and EDA.

PROJECTS (Check out more [here](#))

StructuraAI | Python, OpenAI API, PyMuPDF, JSON Schema, Document Intelligence Analytics | [GitHub](#)

- Built a document intelligence pipeline to extract structured metadata and schema definitions from financial document PDFs into machine-readable JSON outputs.
- Designed parsing logic to capture table layouts, field names, data formats, lengths, and metadata from semi-structured documents and transform them into structured records for downstream analytics workflows.
- Automated document-to-data conversion to support scalable reference-data extraction, improve consistency, and reduce manual processing effort.

KaikoData-Agent | Python, OpenAI Function Calling, Kaiko Market Data API | [GitHub](#)

- Built an agentic AI-powered cryptocurrency market data assistant that converts natural-language trading and market data requests into structured Kaiko API calls using OpenAI function calling.
- Integrated Kaiko Market Data APIs to retrieve OHLCV and tick-level cryptocurrency trading data and export datasets in CSV/JSON formats for quantitative analysis and financial research workflows.
- Automated market data acquisition, formatting, and delivery workflows to reduce manual query effort and support repeatable downstream analysis.

AgenticRAG | Python, OpenAI Function Calling, RAG, Vector Search, ChromaDB, Embeddings | [GitHub](#)

- Built an end-to-end agentic RAG application that uses OpenAI function calling and local vector retrieval to answer Python programming questions with grounded context from official documentation.
- Constructed a local documentation knowledge base using token-based chunking and embeddings to support semantic retrieval across tutorials, reference material, and concept explanations.
- Designed and orchestrated multi-step retrieval workflows combining semantic search, documentation lookup, and context selection to improve response relevance, reduce hallucinations and increase reliability of technical answers.

SKILLS

- Programming:** Python, SQL, PySpark
- Data Platforms:** Databricks, Databricks Notebooks, Unity Catalog, SQL Warehouses, Azure, ADLS, GCP, BigQuery, AWS S3, MongoDB, PostgreSQL
- Data Engineering:** ETL, ELT, Data Pipelines, Data Validation, Data Integration, Data Quality, Pipeline Orchestration, Airflow, Apache Spark, Kafka
- AI Engineering:** LLMs, RAG, OpenAI API, Prompt Engineering, Embeddings, Vector Search, ChromaDB, Tool Calling, Structured Outputs, Agentic Workflows
- Analytics & Product:** Product Analytics, KPI Definition, Metric Definitions, Dashboard Development, Power BI, Data Visualization, User Analytics
- Engineering Practices:** Git, Jira, Confluence, CI/CD, SDLC, Agile Development, Object-Oriented Programming, Stakeholder Collaboration, Requirements Gathering